

Project 1 – Milestone 2

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Introduction:

In Milestone 1, three classification algorithm such as Decision Tree (CART via rpart), PART, and RIPPER (JRip) were evaluated on the German Credit dataset using repeated 10-fold cross-validation and statistical comparison through one-way analysis of variance (ANOVA). The results indicated no statistically significant difference in mean classification accuracy among the models under clean data conditions.

The goal of Milestone 2 is two-fold. Firstly, to assess the robustness of each classifier in the presence of noisy labels by adding 10% random class label noise. Secondly, to identify the top-performing classifier on the basis of four parameters: rule perspicacity, accuracy, stability, and robustness.

Task 4: Robustness to Noise

Method:

To test robustness, 10% of the examples in the dataset were chosen at random, and their class labels were flipped, so that good became bad and vice versa. This created a noisy dataset with 100 incorrect labels among 1,000 examples.

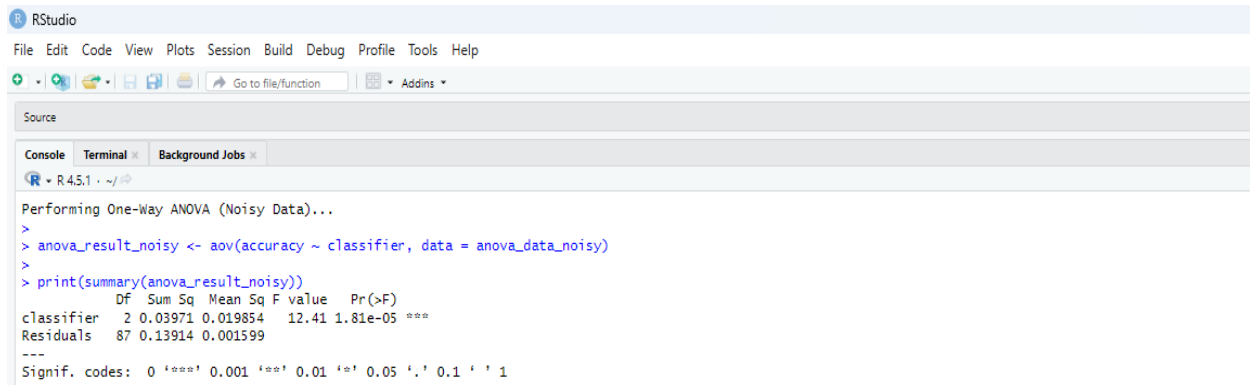
The same repeated 10-fold cross-validation procedure used in Milestone 1 was applied:

- 10 folds
- 3 repeats
- Total resamples per classifier: 30

The accuracy was measured for each resampling. Furthermore, One-way ANOVA was performed to check for the existence of any significant differences among the classifiers when working under noisy conditions. When there were significant differences, Tukey's Honestly Significant Difference (HSD) test was used.

Results:

The one-way ANOVA revealed a statistically significant difference in mean accuracy among classifiers:



```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
R 4.5.1 ~
Performing One-Way ANOVA (Noisy Data)...
> anova_result_noisy <- aov(accuracy ~ classifier, data = anova_data_noisy)
> print(summary(anova_result_noisy))
              Df Sum Sq Mean Sq F value    Pr(>F)
classifier     2  0.03971  0.019854   12.41 1.81e-05 ***
Residuals    87  0.13914  0.001599
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$F(2, 87) = 12.41, p < .001$

This indicates that at least one classifier performed significantly differently under noise conditions.

Mean classification accuracies were:



```
> # Compare Noise Impact
> cat("\nMean Accuracy (Noisy Data):\n")
Mean Accuracy (Noisy Data):
> print(tapply(accuracy_noisy, classifier, mean))
DecisionTree      PART      RIPPER
0.6656729      0.6223147      0.6679819
>
```

- Decision Tree: 0.666
- PART: 0.622
- RIPPER: 0.668

Tukey's HSD post hoc test indicated:

- PART performed significantly worse than Decision Tree ($p < .001$).
- PART performed significantly worse than RIPPER ($p < .001$).
- No significant difference was found between Decision Tree and RIPPER.

Thus, PART was significantly less robust to label noise, whereas Decision Tree and RIPPER maintained comparable performance.

Interpretation:

Adding label noise decreased the total classification accuracy of all models but the extent of the reduction varied. RIPPER had the best mean accuracy in the presence of noise with Decision Tree coming next. The decline of performance in PART was significantly higher. These results indicate that rule-based sequential covering RIPPER and tree-based partitioning (Decision Tree) perform better with moderate amounts of corruption of the class-labels compared to the hybrid rule-induction strategy of PART.

Task 5: Comparative Evaluation and Final Model Selection

Overview:

In this project milestone, three classifiers were evaluated on the German Credit dataset: Decision Tree (CART), PART, and RIPPER (JRip). The models were compared using four criteria:

- i. Rule perspicacity (how simple and clear the rules are)
- ii. Mean accuracy
- iii. Stability (consistency across cross-validation)
- iv. Robustness to noise (performance after adding 10% label noise)

Each classifier was ranked from 1 (best) to 3 (worst) for each factor. The classifier with the lowest average score was selected as the best overall model.

1. Rule Perspicacity:

RIPPER generated only three rules, thus it is the most interpretable and concise model. The rule base was simple, with two explicit rules for the “bad” class and a default rule for the “good” class.

This simple structure improves the clarity and interpretability of the model. The Decision Tree generated a moderately complex model with multiple splits. Although interpretable in its hierarchical structure, it was necessary to trace the paths to generate rules. PART generated 51 rules, thus having a relatively complex model with a less parsimonious rule base. Although locally interpretable, the large number of rules decreases clarity and perspicacity.

Ranking (Rule Perspicacity):

- RIPPER = 1
- Decision Tree = 2
- PART = 3

2. Mean Accuracy:

Under the original dataset (without noise), no statistically significant differences were observed among classifiers, as indicated by a one-way ANOVA ($p = .396$). Therefore, accurate performance was considered comparable. Under 10% class-label noise, statistically significant differences emerged ($F(2, 87) = 12.41, p < .001$). Tukey's HSD test indicated:

- RIPPER significantly outperformed PART ($p < .001$).
- Decision Tree significantly outperformed PART ($p < .001$).
- No significant difference between RIPPER and Decision Tree.

Mean accuracies under noise were:

- RIPPER: 0.668
- Decision Tree: 0.666
- PART: 0.622

RIPPER achieved the highest mean accuracy overall, followed closely by Decision Tree.

Ranking (Mean Accuracy):

- RIPPER = 1
- Decision Tree = 2
- PART = 3

3. Stability:

Stability was measured by examining the variability across 30 resamples of cross-validation. The Decision Tree had stable results across the folds, with only slight changes in accuracy. RIPPER had stable accuracy levels, even in the presence of noise, which clearly depicts excellent generalization performance. PART had higher variability across the folds and a substantial decrease in accuracy with the presence of noise.

Ranking (Stability):

- RIPPER = 1
- Decision Tree = 2
- PART = 3

4. Robustness to Noise:

Robustness was tested by comparing the results before and after injecting 10% class-label noise. RIPPER performed with little change in accuracy, retaining the highest mean accuracy among the classifiers even under noisy conditions. Decision Tree was also robust, with only a slight drop in accuracy. PART had a large drop in accuracy and was statistically worse than RIPPER and Decision Tree even under noise.

Ranking (Robustness to Noise):

- RIPPER = 1
- Decision Tree = 2
- PART = 3

5. Overall Ranking:

The rankings for each criterion are summarized in Table 1.

Classifier	Rule Perspicacity	Accuracy	Stability	Robustness	Mean Rank
RIPPER	1	1	1	1	1.00
Decision Tree	2	2	2	2	2.00
PART	3	3	3	3	3.00

RIPPER achieved the lowest (best) average ranking across all four evaluation criteria.

Conclusion:

According to experimental results, RIPPER is the best classifier to use in this dataset. It has the most concise and interpretable rule base, has the greatest mean predictive accuracy, is very stable in cross-validation folds and is also better robust when there is label noise.

Despite the fact that the Decision Tree was a competitive tool, and it can be chosen in case hierarchical interpretability is needed, RIPPER offers the most appropriate compromise between interpretability, predictive performance, stability, and robustness. Although PART, being rule-based and expressive was less robust and stable compared to the other classifiers.

Thus, RIPPER is chosen as the ideal general classifier to German Credit data set in the given research.